LOW LEVEL DESIGN

Soil sensor

Revision B · 2026-11-05

Made by trace from the project's records at commit `a41e9d0c2b17`

Every figure and value in this document names the record it came from. A value marked pending has not been worked out yet.

Revisions

Revision	Date	Commit	What changed
A	2026-10-21	`9c02e1f`	First release
B	2026-11-05	`a41e9d0`	Released

Contents

1 Modules . . . 1
2 Pin plan . . . 1
3 Power tree . . . 2
4 Power budget . . . 2
5 Thermal . . . 2
6 Loads and ratings . . . 2
7 Interfaces . . . 2
8 Firmware . . . 2
9 Board . . . 2
10 Mechanical . . . 2
11 Verification . . . 2
12 Glossary . . . 3

1 Modules

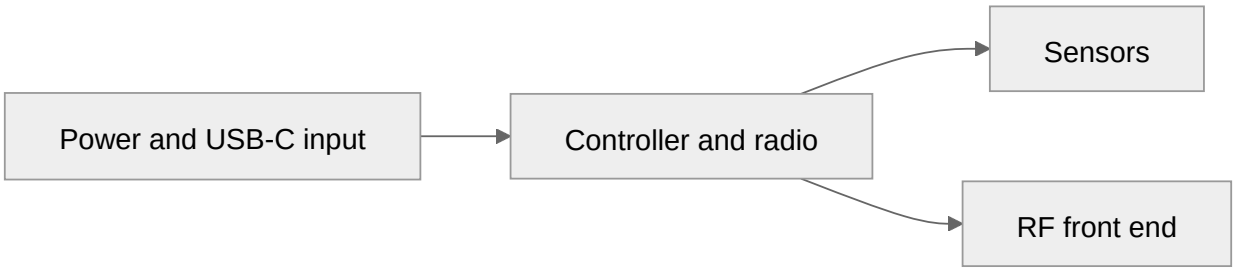


Figure 1: The four schematic sheets and how power and signals pass between them.

2 Pin plan

Pending: Every used pin: net, function and peripheral, and the boot and debug pins.

3 Power tree

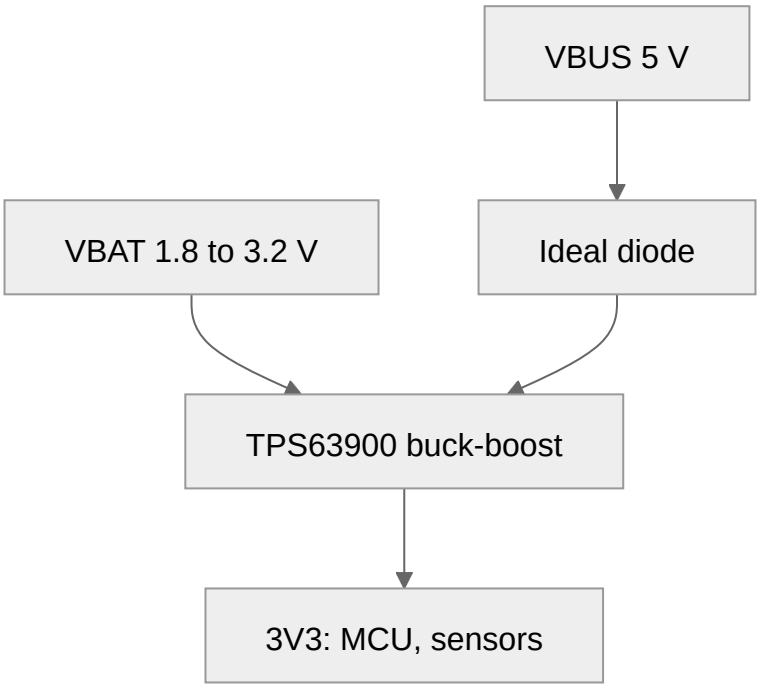


Figure 2: Every rail and what it feeds.

4 Power budget

Pending: Current per rail at typical and worst case, regulator headroom, battery life where it applies.

5 Thermal

Pending: The hottest parts, temperature rise and margin to each part's rating.

6 Loads and ratings

Pending: Connector and track current against ratings, bus loading.

7 Interfaces

Pending: Each interface: connector, signals, levels and the protocol specification section that covers it.

8 Firmware

Pending: Structure, start-up and main loop, and how each interface is driven.

9 Board

Pending: Stackup, placement, routing rules and the fabrication choices, with the board renders.

10 Mechanical

Pending: The case: dimensions, clearances, mounting and cut-outs, and how it assembles.

11 Verification

Pending: What was simulated and tested, the results, and which requirement each proves.

12 Glossary

Pending: Every acronym and term used in this document.